

Technology Stack

2. Requirement Analysis | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
Credit Card Approval Prediction	Ankitha Vana	Finance	July 1, 2026

Technology Overview

The solution uses Python-based data science and web development tools for analysis, model building, visualization, and deployment.

Tools and Libraries

Technology	Purpose in Project
Anaconda Navigator	Manage Python environments and launch Jupyter Notebook for data analysis.
PyCharm / VS Code	Write, debug, and organize the Flask application code.
Python 3.8+	Primary programming language for data processing, model training, and web application development.
NumPy	Numerical computation and array operations.
Pandas	Read CSV files, clean records, merge datasets, and manipulate tabular data.
Matplotlib and Seaborn	Create univariate, multivariate, and descriptive visualizations.
Scikit-learn	Preprocessing, train-test split, model training, metrics, and model selection.
XGBoost	Gradient boosting classifier for high-performance structured-data prediction.
Flask	Lightweight web framework used to serve the prediction form and result page.
IBM Watson Machine Learning	Optional cloud deployment pipeline for real-time scoring endpoints.

Machine Learning Algorithms

- Logistic Regression: interpretable baseline classifier for binary approval prediction.
- Decision Tree: rule-based model useful for understanding feature splits.
- Random Forest: ensemble of trees that improves stability and reduces overfitting.
- XGBoost: gradient boosting model selected for strong performance on tabular classification problems.